

Safety Constraint Module (SCM)

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Canonical Definition

Safety Constraint Module (SCM) defines the structural conditions under which mandatory safety constraints are established, enforced, monitored, and preserved throughout autonomous operation.

SCM governs safety constraints.

The module establishes the governance conditions required to ensure that autonomous systems continuously operate within approved safety limits, preventing unacceptable operational risk to people, assets, infrastructure, and the environment.

Safety cannot depend on autonomous discretion alone.

Safety requires governed constraints.

SCM governs those constraints.

Module Operational Role

SCM defines the safety constraint layer of ACS.

Its role is to preserve operational safety by ensuring that autonomous execution continuously respects mandatory safety constraints.

Module Operational Space

SCM governs:

- safety constraints,
 - operational safety limits,
 - autonomous safety governance,
 - safety boundary enforcement,
 - operational hazard prevention,
 - governed safety controls,
 - safety compliance,
 - continuous safety protection.
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Module Function

The module applies wherever organizations must establish and enforce mandatory safety constraints before and during autonomous operations.

Minimum Implementation Framework

1. Define the Safety Constraint Object Identify which safety constraints are mandatory for autonomous operation.

2. Define Safety Conditions

Establish governance conditions governing operational safety limits and acceptable safety thresholds.

3. Define Safety Failure Logic

Identify conditions where safety constraints become ineffective, violated, bypassed, or governance-invalid.

4. Define Governance Response

Define governance actions for safety intervention, execution restriction, emergency shutdown, escalation, or recovery.

5. Preserve Safety Traceability

Preserve safety constraints, governance decisions, operational evidence, detected safety events, and supporting documentation.

Use Case 1 — Autonomous Passenger Shuttle

Scenario

An autonomous shuttle transports passengers through a public transportation network.

Application

SCM continuously enforces passenger safety constraints throughout autonomous operation.

Result

Passenger transport remains safe, controlled, and continuously governed.

Use Case 2 — Autonomous Industrial Robot

Scenario

An autonomous manufacturing robot operates alongside human workers.

Application

SCM enforces mandatory safety constraints whenever human presence or hazardous operating conditions are detected.

Result

The organization preserves worker safety while maintaining governed autonomous operation.

Canonical Closing Statement

Trustworthy autonomy depends on non-negotiable safety limits. Safety Constraint ensures that autonomous systems continuously operate within governed safety boundaries, protecting people, assets, and operational environments throughout the complete autonomous operational lifecycle.

Related Documents

[→ Autonomous Constraint Standard \(ACS\)](#)

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Canonical Source

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